

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 209 — Past Questions & Materials
by Renance

Course: MTH 209
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

This study pack was compiled and restructured by Renance for MTH 209 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 2 — MTH 209 2018 2019

Level 200 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
B.Sc. (MATHEMATICS) DEGREE FIRST SEMESTER EXAMINATION 2018/2019
COLLEGE OF SCIENCE, DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

TIME: 2 HOURS

MTH 219- INTRODUCTION TO OPERATION RESEARCH

INSTRUCTION: ANSWER QUESTION ONE AND ANY OTHER THREE QUESTIONS

1. (a) Using the graphical method

Maximize $Z = 80x + 120y$

Subject to the constraints

$$x + y \leq 8 = (0, 8) (8, 0)$$

$$x \geq 2$$

$$y \geq 3$$

$$20x + 5y \geq 40 = (2, 0) (0, 8)$$

$$x, y \geq 0$$

(b) Minimize $Z = 20x + 10y$

Subject to the constraints

$$x + 2y \geq 4$$

$$3x + y \leq 30$$

$$4x + 3y \geq 24$$

$$x, y \geq 0$$

(c) Use the simplex method to solve the linear programming problem

Maximize $Z = 3x + 5y + 4k$

Subject to the constraints

$$2x + 3y \leq 8$$

$$2y + 5k \leq 10$$

$$3x + 2y + 4k \leq 5$$

$$x, y, k \geq 0$$

2. A manufacture produces two different models: X and Y of the same product. Model X makes a contribution of #50 per unit and Model Y #30 per unit towards total profit. Raw materials r_1 and r_2 are required for production. At least 18kg of r_1 and 12kg of r_2 must be used daily. Also at most 34 hours of labour are to be utilized. A quality of 2kg of r_1 is

needed for model X and 1kg of r_1 for Model Y. for each of X and Y, 1kg of r_2 is required. It takes 2 hours to manufacture Model X and 2 hours to manufacture Model Y. How many units of each Model should be produced to maximize the profit using the graphical method?

3. (a) Write a short note on the following

(i) Feasible solution

(ii) Infeasible solution

(iii) Feasible region

(iv) Optimal solution

(v) Corner point feasible solution

(b) Convert the following primal linear programming model problems to dual linear programming problems.

(i) Maximize $Z = 5x_1 + 12x_2 + 4x_3$

Subject to the constraints

$$x_1 + 2x_2 + x_3 \leq 10$$

$$2x_1 - x_2 + 3x_3 = 8$$

$$x_1, x_2, x_3 \geq 0$$

(ii) Minimize $Z = 15x_1 + 12x_2$

Subject to the constraints

$$x_1 + 2x_2 \geq 3$$

$$2x_1 - 4x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

(iii) Maximize $Z = 5x_1 + 6x_2$

Subject to

$$x_1 + 2x_2 = 5$$

$$-x_1 + 5x_2 \geq 3$$

$$4x_1 + 7x_2 \leq 8$$

$$x_1 \text{ unrestricted, } x_2 \geq 0$$

4. (a) Use the penalty (Big M) method to solve the linear programming problem.

$$\text{Minimize } Z = 5x_1 + 3x_2$$

Subject to the constraints

$$2x_1 + 4x_2 \leq 12$$

$$2x_1 + 2x_2 = 10$$

$$5x_1 + 2x_2 \geq 10$$

$$x_1, x_2 \geq 0$$

- (b) Find the optimal solution using the optimal primal basis.

$$\text{Maximize } Z = 3x_1 + 5x_2$$

Subject to the constraints

$$x_1 + 2x_2 + x_3 = 5$$

$$-x_1 + 3x_2 + x_4 = 2$$

$$x_1, x_2, x_3, x_4 \geq 0$$

5. Solve the linear programming problem by using the two-phase simplex method

$$\text{Minimize } Z = x_1 + x_2$$

Subject to the constraints

$$2x_1 + 4x_2 \geq 4$$

$$x_1 + 7x_2 \geq 7$$

$$x_1, x_2 \geq 0$$

6. (a) A salesman located in a city A decided to travel to city B. He knew the distances of alternative routes from city A to city B. He then drew a highway network map. The city of origin A is city 1. The destination city B is city 10. Other cities through which the salesman will have to pass through are numbered 2 to 9. The routes between cities are in distance as shown below. The salesman's problem is to find the shortest route that covers all the selected cities from A to B. Solve the salesman's problem. Show all workings and put the information in a transition matrix/TPM.

$$d(1,2) = 4, d(2,5) = 7, d(5,8) = 4, d(8,10) = 7,$$

$$d(2,6) = 10, d(6,8) = 3, d(6,9) = 7, d(9,10) = 9,$$

$$d(2,7) = 5, d(7,9) = 4, d(5,9) = 8, d(7,8) = 8,$$

$$d(1,3) = 6, d(3,5) = 3, d(3,6) = 8, d(3,7) = 4,$$

$$d(1,4) = 3, d(4,5) = 6, d(4,6) = 10, d(4,7) = 5$$

- (b) What is the advantage of the simplex method over the graphical method?

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
 B.Sc. (MATHEMATICS) DEGREE FIRST SEMESTER EXAMINATION 2017/2018
 COLLEGE OF SCIENCE, DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
 TIME: 2 HOURS
 MTH 219- INTRODUCTION TO OPERATION RESEARCH
 INSTRUCTION: ANSWER ANY THREE QUESTIONS

1(a) The ABC printing company is facing a tight financial squeeze and is attempting to cut costs wherever possible. At present it has only one printing contract and, luckily the book is selling well in both the hardcover and paperback editions. It has just received a request to print more copies of this book in either the hardcover or paperback form. The printing cost for hardcover books is #600 per 100 while that for paperback is only #500 per 100. Although the company is attempting to economize, it does not wish to lay off any employee. Therefore, it feels obliged to run its two printing presses at least 80 and 60 hours per week respectively. Press 1 can produce 100 hardcover books in 2 hours or 100 paperback books in 1 hour. Press 2 can produce 100 hardcover books in 1 hour or 100 paperbacks books in 2 hours. Using the BIG M METHOD, determine how many books of each type should be printed in order to minimize costs.

(b) Use graphical method to solve the following LP Models

(i) Maximize $Z = 4x + 4y$.

Subject to $x + 2y \leq 10$,
 $6x + 6y \leq 36$,
 $x \leq 6$

$Z = 600x_1 + 500x_2$
 subject to
 $2x_1 + x_2 \geq 80$
 $x_1 + 2x_2 \geq 60$
 $x_1, x_2 \geq 0$

(ii)

Minimize $Z = x - y$.

Subject to the constraint

$-2x + y \leq 8$,

$-x + 2y \leq -24$,

$x, y \geq 0$

Handwritten solution for (ii) using the Big M method:

	C_b	C_j	600	500	0	0	M	M		
			x_1	x_2	s_1	s_2	A_1	A_2	Sol.	Ratio
R_1	M	2	1	-1	0	-1	0	-M	80	80/2 = 40
R_2	M	1	2	0	1	0	1	-M	60	60/2 = 30

$Z_j = 3M \quad 3M \quad -M \quad -M \quad M \quad M \quad 140M$
 $Z_j - C_j = 600M - 3M \quad 500M - 3M \quad -M \quad -M \quad 0 \quad 0$

(c) Write the dual of the LP Model,

Maximize $Z = x - y + 3k$,

subject to the constraint

$x + y + k \leq 10$,

$2x - k \leq 2$,

$2x - 2y - 3k \leq 6$,

$x, y, k \geq 0$

2(ai) Enumerate the steps involve in carry out a successful operation research

(ii) Briefly explain what is a system of operation research and its characteristics?

(b) Use graphical method to solve the LP Model problem

Maximize $Z = 2x + 3y$

subject to the constraint

$$x + y \leq 30,$$

$$y \geq 3, 0 \leq y \leq 12,$$

$$0 \leq x \leq 20,$$

$$x - y \geq 0,$$

$$x, y \geq 0$$

3(a) What are the limitations of Linear Programming? Write short note on the following: slack variable, surplus variable, feasible solution, infeasible solution and corner points.

(b) Use the Simplex method to solve the following LP problem

$$\text{Maximize } Z = 3x + 5y + 4k,$$

Subject to the constraints,

$$2x + 3y \leq 8,$$

$$2y + 5k \leq 10,$$

$$3x + 2y + 4k \leq 15,$$

$$x, y, k \geq 0$$

(ai) Enumerate four special cases in Simplex method

(i) Write short note on Duality

(ii) State the three properties of an LP model.

(b) XYZ Company has 3 departments-Assembly, Painting and Packing, and can make three types of almirahs. An almirah of type 1 requires one hour of assembly, 40 minutes of painting and 20 minutes of packing time respectively. Similarly an almirah of type 11 needs 80 minutes, 20 minutes and one hour respectively. The almirah of type 111 requires 40 minutes each of assembly, painting and packing time. The total time available at assembly, painting and packing department is 600 hours, 400 hours and 800 hours respectively. Determine the number of each type of almirahs that should be produced in order to maximize the profit. The unit profit for types 1, 11, and 111 is \$40, \$80 and \$60 respectively.

Suppose that the manager of XYZ Company is thinking of renting the production capacities of the three departments to another almirah the manufacturer- ABC Company. ABC Company is interested in minimizing the rental charges. On the other hand, the XYZ Company would like to know the worth of production hours to them, in each of the departments to determine the rented rates

(i) Formulate this problem as an LP problem and solve it to determine the number of each type of almirahs that should be produced by the XYZ Company in order to maximize its profits.

(ii) For LP problem in (a) formulate its dual and interpret your results.

Paper 2 of 2 — MTH 209 MATERIAL

Level 200 • First Semester • Handout

This source paper is a scanned document; its pages are reproduced below.

4

236

$$x_1 - 3x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

starting point (2, 0)

4.9. Solve by gradient projection:

$$\text{minimize } z = 2(x_1 - 4)^2 + (x_2 - 4)^2$$

$$\text{subject to } 2x_1 + 3x_2 \leq 12$$

$$x_1 + x_2 \leq 2$$

$$3x_1 - 4x_2 \leq 6$$

$$x_1 \geq 0$$

starting point (0, 2)

4.10. For the problem

$$\text{maximize } z = 6(x_1 - 2)^2 + 4(x_2 - 2)^2$$

$$\text{subject to } 0 \leq x_1 \leq 4$$

$$0 \leq x_2 \leq 4$$

solve by using gradient projection starting at the point (2, 2). Replace x_1 by $4 \sin^2 y_1$ and x_2 by $4 \sin^2 y_2$. Note by this transformation the resulting problem is unconstrained; solve by differential calculus. Compare the solutions generated by each method.

4.11. Consider the problem

$$\text{maximize } z = 4x_1 + 7x_2$$

$$\text{subject to } x_1 + 2x_2 \leq 8$$

$$7x_1 + 4x_2 \leq 28$$

$$x_1, x_2 \geq 0$$

By introducing an additional variable x_3 , the problem may be recast as

$$\text{maximize } z = x_3$$

$$\text{subject to } x_1 + 2x_2 \leq 8$$

$$7x_1 + 4x_2 \leq 28$$

$$x_3 - 4x_1 - 7x_2 = 0$$

$$x_1, x_2 \geq 0$$

Show that both problems have the same solution by solving each, using the simplex algorithm.

5

QUADRATIC PROGRAMMING

5.1 INTRODUCTION

A quadratic programming problem is one in which a quadratic objective function is to be optimized. The constraints are assumed to be linear inequalities, and the variables in the problem must be nonnegative.

If all the functions in the model are linear, the methods described in Chapter 2 are suitable for solving the problem. If the constraints are not linear or if the objective function is neither linear nor quadratic, the problem is a nonlinear programming problem to which the material in Chapters 3 and 4 is relevant.

The emphasis here will not be on the algorithms that have been conceived for quadratic programming but on the Lagrangian multiplier method. This derivation produces the Kuhn-Tucker conditions for quadratic programming, using calculus. This approach is intended to provide a better understanding of the optimality conditions for quadratic programming and to give some insight into the interpretation of the Lagrangian multipliers. This point is of particular significance because so many applications of quadratic programming appear in the economics, management science, and operations research literature. Some of the implied quadratic programming problems and models are often overlooked. Examples of such cases occur when imperfectly competitive output markets are assumed and only nonnegative levels of inputs, outputs, and prices are permitted. These assumptions cannot be introduced into programming models without linear inequalities or more complex inequality constraints.

237

238

5

In this chapter all functions are assumed to be differentiable, and all variables take only real values (imaginary numbers are excluded). This is required since derivatives will be applied to find the optimality conditions.

Consider the constraints of the linear programming problem presented in Example 2.9:

$$4x_1 + x_2 \leq 35 \quad x_1 + 4x_2 \leq 20 \quad x_1, x_2 \geq 0$$

The feasible area determined by these constraints is the polygon $OABC$ in Figure 5.1.

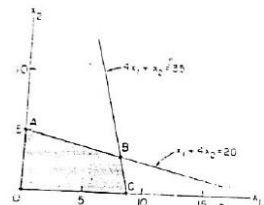


Fig. 5.1. Feasible area determined by constraints.

An optimal solution to a linear programming problem occurs at one of the corner points of the feasible polygon. In Example 2.9, $z = 10x_1 + 15x_2$ has been maximized, subject to the constraints that form the polygon in Figure 5.1. The maximum value of z , 125, occurs at $x_1 = 8$ and $x_2 = 3$, which is at point B .

The optimal solution to a quadratic programming problem does not necessarily occur at a point on the perimeter of the feasible polygon. Suppose the example presented above is changed by adding two quadratic terms, $-1.25x_1^2$ and $-3.75x_2^2$, to the objective function. Then the problem is to find x_1 and x_2 to

$$\begin{aligned} \text{maximize } w &= 10x_1 + 15x_2 - 1.25x_1^2 - 3.75x_2^2 \\ \text{subject to } 4x_1 + x_2 &\leq 35 \\ x_1 + 4x_2 &\leq 20 \\ x_1, x_2 &\geq 0 \end{aligned}$$

The maximum value of w when the quadratic function is unconstrained

QUADRATIC PROGRAMMING

239

is $w = 35$, which occurs when $x_1 = 4$ and $x_2 = 2$; the first-order (necessary) conditions are

$$\begin{aligned} \frac{\partial w}{\partial x_1} &= 10 - 2.5x_1 = 0 & x_1 &= 4 \\ \frac{\partial w}{\partial x_2} &= 15 - 7.5x_2 = 0 & x_2 &= 2 \end{aligned}$$

and the required second-order (sufficient) conditions are satisfied at $x_1 = 4$ and $x_2 = 2$.

Furthermore, $x_1 = 4$ and $x_2 = 2$ satisfies each of the inequalities

$$4x_1 + x_2 < 35 \quad x_1 + 4x_2 < 20 \quad x_1, x_2 > 0$$

Therefore, $x_1 = 4$ and $x_2 = 2$ is the optimal solution to the quadratic programming problem defined in the previous paragraph. This solution is not on the boundary of the feasible polygon since the inequality is strictly satisfied for each constraint. This is not often the case, as will be shown in Section 5.2, and more sophisticated methods are needed to solve the problem.

5.2 METHOD OF LAGRANGIAN MULTIPLIERS

Example 5.1

The method of Lagrangian multipliers for quadratic programming will be introduced by considering a simple example. Suppose that a firm produces two commodities in quantities x and z . The prices are assumed to be P_x and P_z , which define the respective average revenue functions for x and z :

$$P_x = 10 - x + 150/z \quad P_z = 64 - 8z + 250/z$$

respectively. It costs \$200 to produce a unit of x and \$100 to produce a unit of z . The firm's budget is \$4700. Therefore, $200x + 100z \leq 4700$, or $2x + z \leq 47$, is the firm's budget constraint.

Suppose that the firm attempts to maximize its total revenue TR .

$$TR = xP_x + zP_z = 10x - x^2 + 64z - 8z^2 + 400$$

240

5

The optimization problem is

$$\begin{aligned} &\text{maximize } TR = 10x - x^2 + 64z - 8z^2 + 400 \\ &\text{subject to } 2x + z \leq 47 \\ &\quad x, z \geq 0 \end{aligned}$$

This problem can be solved using the Lagrangian multiplier technique. The procedure is the same as the approach to linear programming described in Chapter 2 and as that to nonlinear models in Chapter 3.

The budget constraint is converted to an equation by adding the slack variable y ; $2x + z + y = 47$. Each variable is set equal to the square of a real value; $x = v^2$, $z = w^2$, and $y = u^2$. Assuming that v , w , and u have real values implies that v^2 , w^2 , and u^2 are nonnegative. Therefore, x , z , and y are nonnegative values.

The Lagrangian function for this problem is

$$L = 10x - x^2 + 64z - 8z^2 + 400 + \lambda(2x + z + y - 47) + \rho(x - v^2) + \phi(z - w^2) + \pi(y - u^2)$$

This function includes four Lagrangian multipliers λ , ρ , ϕ , and π . One Lagrangian multiplier precedes each constraint in the problem, since there is a budget constraint and three nonnegative variables.

The first-order conditions for this problem are derived by determining ten partial derivatives and setting each equal to zero. These derivatives follow:

$$\frac{\partial L}{\partial x} = 10 - 2x + 2\lambda + \rho = 0$$

$$\frac{\partial L}{\partial z} = 64 - 16z + \lambda + \phi = 0$$

$$\frac{\partial L}{\partial \lambda} = 2x + z + y - 47 = 0$$

$$\frac{\partial L}{\partial y} = \lambda + \pi = 0 \quad \lambda = -\pi$$

$$\frac{\partial L}{\partial \rho} = x - v^2 = 0 \quad x = v^2$$

$$\frac{\partial L}{\partial v} = -2\rho v = 0 \quad \rho v^2 = 0$$

$$\left. \begin{aligned} x = v^2 \\ \rho v^2 = 0 \end{aligned} \right\} x\rho = 0$$

QUADRATIC PROGRAMMING

241

$$\left. \begin{aligned} \frac{\partial L}{\partial \phi} = z - w^2 = 0 \quad z = w^2 \\ \frac{\partial L}{\partial w} = -2\phi w = 0 \quad \phi w^2 = 0 \\ \frac{\partial L}{\partial \pi} = y - u^2 = 0 \quad y = u^2 \\ \frac{\partial L}{\partial u} = -2\pi u = 0 \quad \pi u^2 = 0 \end{aligned} \right\} \begin{aligned} z\phi &= 0 \\ y\pi &= 0 \end{aligned}$$

These results can be reduced to a set of three linear equations in six unknowns with three additional conditions. The variable π can be replaced by minus λ , and the equations that remain are

$$-2x + 2\lambda + \rho = -10$$

$$-16z + \lambda + \phi = -64$$

$$2x + z + y = 47$$

$$x\rho = 0 \quad z\phi = 0 \quad y\lambda = 0$$

Each possibility that the side conditions provide is a set of three linear equations in three unknowns. There are eight of these sets, $2^3 = 8$, since there are three side conditions and each has two variables.

The eight possible cases are indicated in Table 5.1. It should be noted that these are an exhaustive set of possibilities. Some of these eight provide infeasible solutions or solutions that are not unique.

Table 5.1. Possible Solutions

Case	x	z	y	ρ	π	λ	TR
1	0	0	0			0	
2	0	0	0		0		
3	0	0	0		0	0	
4	0	0	0		0	0	
5	0	0	0	0			
6	0	0	0	0	0		
7	0	0	0	0	0	0	
8	0	0	0	0	0	0	

Each case from Table 5.1 is solved below and the results are summarized in Table 5.2. Case 8 provides the optimal solution.

344

The cases in which there is no unique, feasible solution are indicated by an asterisk in Table 5.2.

Case 1: $x = z = y = 0$ no solution
 $2\lambda + \rho = -10$
 $\lambda + \phi = -64$
 $0 = 47$

Case 2: $x = z = \lambda = 0$ $TR = 400$
 $\rho = -10$
 $\phi = -64$
 $y = 47$

Case 3: $x = y = \phi = 0$ $TR = -14264$
 $2\lambda + \rho = -10$ $\rho = -1386$
 $-16z + \lambda = -64$ $\lambda = 688$
 $z = 47$ $z = 47$

Case 4: $x = \phi = \lambda = 0$ $TR = 528$
 $\rho = -10$ $z = 4$
 $-16z = -64$ $y = 43$
 $z + y = 47$ $\rho = -10$

Case 5: $z = y = \rho = 0$ $TR = 82.75$
 $-2x + 2\lambda = -10$ $\lambda = 19.5$
 $\lambda + \phi = -64$ $\phi = 83.5$
 $2x = 47$ $x = 23.5$

Case 6: $z = \rho = \lambda = 0$ $TR = 425$
 $-2x = -10$ $x = 5$
 $\phi = -64$ $\phi = -64$
 $2x + y = 47$ $y = 37$

Case 7: $y = \rho = \phi = 0$ $TR = 289$
 $-2x + 2\lambda = -10$ $x = 21$
 $-16z + \lambda = -64$ $z = 5$
 $2x + z = 47$ $\lambda = 16$

Case 8: $\rho = \phi = \lambda = 0$ $TR = 553$
 $-2x = -10$ $x = 5$
 $-16z = -64$ $z = 4$
 $2x + z + y = 47$ $y = 33$

Only the first-order conditions have been considered for this example. The general quadratic programming problem and the required conditions are presented in Section 5.4.

QUADRATIC PROGRAMMING

Table 5.2. Solutions

Case	x	z	y	ρ	ϕ	λ	TR
1	0	0	0	*	*	*	*
2	0	0	47	-10	-64	0	400
3	0	47	0	-1386	0	688	-14,264
4	0	4	43	-10	0	0	528
5	23.5	0	0	0	83.5	19.5	82.75
6	5	0	37	0	-64	0	425
7	21	5	0	0	0	16	289
8	5	4	33	0	0	0	553

5.3 QUADRATIC FORMS

A quadratic form of the variables $x_1, x_2, \dots, x_m$ is a polynomial in which each term is quadratic. This means x_i^2 or $x_i x_j$ must appear in each term. The equation

$$W = \sum_{j=1}^m \sum_{k=1}^m r_{jk} x_j x_k$$

is a general quadratic form that could also be written as

$$W = \sum_{j=1}^m r_{jj} x_j^2 + \sum_{j=1}^m \sum_{k=1}^m r_{jk} x_j x_k$$

Employing matrix notation, a matrix of the quadratic form is expressed as $W = \mathbf{x}'\mathbf{R}\mathbf{x}$, where $\mathbf{x}$ is an $m \times 1$ vector, $\mathbf{x}'$ is the transpose of the $\mathbf{x}$ vector, and $\mathbf{R}$ is an $m \times m$ matrix. If $\mathbf{R}$ is not a symmetric matrix, one exists that can be substituted for $\mathbf{R}$ to provide the same value for W .

Example 5.2

For example, let

$$W = [x_1, x_2, x_3] \begin{bmatrix} 3 & 2 & 2 \\ 4 & 5 & 2 \\ 5 & 3 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Then

$$W = 3x_1^2 + 5x_2^2 + 7x_3^2 + 6x_1x_2 + 7x_1x_3 + 5x_2x_3$$

$$[x_1 \ x_2 \ x_3]$$

However, the same result is obtained by letting

$$W' = [x_1, x_2, x_3] \begin{bmatrix} 3 & 3 & 3.5 \\ 3 & 5 & 2.5 \\ 3.5 & 2.5 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Comparing the elements of the two coefficient matrices shows that the two matrices have the same main diagonals. Furthermore, the average of the two elements in row i , column j and row j , column i of the nonsymmetric matrix appears in both these locations in the symmetric matrix. For example, in the first matrix the elements in row 1, column 3 and row 3, column 1 are 2 and 5 respectively. In the second case, where the symmetric coefficient matrix is employed, both values are 3.5.

In general, if R is a nonsymmetric square matrix, a square symmetric matrix R^* can be found to provide the same quadratic form, using two rules: (1) the elements of the main diagonal of R become the elements of the main diagonal of R^* and (2) the elements from row i , column j and row j , column i from R are averaged, and this average becomes the element in both row i , column j and row j , column i of R^* .

Definite and Semidefinite Forms

If a quadratic programming problem has an objective function of $w = c'x + x'Rx$, sometimes the quadratic form $x'Rx$ will enable us to consider very simple second-order optimizing conditions depending on determinants formed from within the matrix R . Let R_i be the determinant of an $i \times i$ matrix. The determinant R_1 is formed by ignoring all rows and columns except the first element of the matrix R . The determinant R_2 is formed using only the first two rows and first two columns of the matrix R .

$$R_1 = |r_{11}|$$

$$R_2 = \begin{vmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{vmatrix}$$

$$R_3 = \begin{vmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{vmatrix}$$

$$\vdots$$

$$R_n = |R|$$

The quadratic form $x'Rx$ is negative definite if $(-1)^{i+1} R_i < 0$, $i = 1, \dots, m$; $R_1 < 0$, $R_2 > 0$, $R_3 < 0, \dots$. If $(-1)^{i+1} R_i \leq 0$ and at least one of $(-1)^{i+1} R_i \neq 0$, the quadratic form $x'Rx$ is negative semidefinite. If $R_i > 0$, $i = 1, \dots, m$, the quadratic form is positive definite. If $R_i \geq 0$, $i = 1, \dots, m$, and at least one of $R_i \neq 0$, the quadratic form is positive semidefinite.

Variance and Covariance

One quadratic form of particular interest in quantitative economic models is the variance-covariance form. This will be needed to study portfolio models in the final section of this chapter.

Suppose that C is a matrix having m rows and n columns with elements c_{ij} , $i = 1, \dots, m$, and $j = 1, \dots, n$. If t_j is the sum of the elements in column j , then t_j/m is the average or mean of the elements in column j ; let $u_j = t_j/m$.

Example 5.3

If

$$C = \begin{bmatrix} 3 & 2 & 8 \\ 2 & 5 & 3 \\ 5 & 1 & 4 \\ 6 & 0 & 5 \end{bmatrix}$$

then $t_1 = 16$, $t_2 = 8$, $t_3 = 20$, $u_1 = 4$, $u_2 = 2$, and $u_3 = 5$.

The sample variance of the elements in column j of the matrix C is denoted as s_j^2 , where

$$s_j^2 = \frac{\sum_{i=1}^m (c_{ij} - u_j)^2}{(n - 1)}$$

For the first two columns of the matrix C ,

$$s_1^2 = \frac{(3-4)^2 + (2-4)^2 + (5-4)^2 + (6-4)^2}{4-1} = \frac{10}{3}$$

and

$$s_2^2 = \frac{(2-2)^2 + (5-2)^2 + (1-2)^2 + (0-2)^2}{4-1} = \frac{14}{3}$$

The sample covariance between the i th and j th columns of C is denoted as s_{ij} and defined by

$$s_{ij} = \frac{\sum_{k=1}^n (c_{ki} - \bar{c}_i)(c_{kj} - \bar{c}_j)}{(n-1)}$$

For the first two columns of C

$$s_{12} = \frac{(3-4)(2-2) + (2-4)(5-2) + (5-4)(1-2) + (6-4)(0-2)}{4-1} = -\frac{11}{3}$$

For the three columns of the matrix C there are three variances s_{11}^2 , s_{22}^2 , and s_{33}^2 and three covariances s_{12} , s_{13} , and s_{23} . These can all be represented in a symmetric matrix as

$$\begin{bmatrix} s_{11}^2 & s_{12} & s_{13} \\ s_{12} & s_{22}^2 & s_{23} \\ s_{13} & s_{23} & s_{33}^2 \end{bmatrix} = \begin{bmatrix} 10/3 & -11/3 & 0 \\ -11/3 & 14/3 & -5/3 \\ 0 & -5/3 & 14/3 \end{bmatrix}$$

The results are the matrix product of the quadratic form $(C' \cdot C)/(n-1)$, where the matrix C is formed by subtracting the column mean from each element of the respective column.

$$\hat{C} = \begin{bmatrix} (3-4) & (2-2) & (8-5) \\ (2-4) & (5-2) & (3-5) \\ (5-4) & (1-2) & (4-5) \\ (6-4) & (0-2) & (5-5) \end{bmatrix} = \begin{bmatrix} -1 & 0 & 3 \\ -2 & 3 & -2 \\ 1 & -1 & -1 \\ 2 & -2 & 0 \end{bmatrix}$$

Example 5.4

Suppose that p and q represent vectors of n prices and commodities respectively, so that

$$E = \sum_{i=1}^n p_i q_i = p'q$$

is the quadratic form of the expenditures on the n items. If the actual prices are unknown, the average price vector $\bar{p}$ may be used as an estimator for p , and the expected or average expenditure

QUADRATIC PROGRAMMING

ture E would be

$$E = \sum_{i=1}^n \bar{p}_i q_i = \bar{p}'q$$

Here $\bar{p}_i$ represents the average of observations on the price of commodity i . The variance of the expenditures $V(E)$ can also be specified if the variances and covariances among the prices are known. If σ_{ii}^2 is the variance of p_i and σ_{ij} is the covariance between p_i and p_j ,

$$V(E) = \sum_{i=1}^n \sigma_{ii}^2 q_i^2 + \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} q_i q_j = q'Sq$$

where

$$S = \begin{bmatrix} s_{11}^2 & s_{12} & \cdots & s_{1n} \\ s_{12} & s_{22}^2 & \cdots & s_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{n1} & s_{n2} & \cdots & s_{nn}^2 \end{bmatrix}$$

is a symmetric variance-covariance matrix for the stock prices. The matrix form of the variance of expenditures is a quadratic form in the vector of variables in q .

In portfolio models the most common objective function is a combination of the average expenditures E and the variance among the expenditures $V(E)$. Some of these models are presented in Section 5.8.

*5.4 GENERAL QUADRATIC PROBLEM

In matrix notation the quadratic programming problem is

$$\text{optimize } w(x) = c'x + x'Rx \quad (5.1)$$

$$\text{subject to } Ax \leq b \quad (5.2)$$

$$x \geq 0 \quad (5.3)$$

248

where

$$\begin{aligned}
 c' &= [c_1, c_2, \dots, c_m] \\
 x' &= [x_1, x_2, \dots, x_m] \\
 b' &= [b_1, b_2, \dots, b_n] \\
 A &= \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix} \quad R = \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1m} \\ r_{21} & r_{22} & \dots & r_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \dots & r_{mm} \end{bmatrix}
 \end{aligned}$$

As for the linear case let $y_i, i = 1, \dots, n$, be defined as a slack variable so that $y' = [y_1, y_2, \dots, y_n], y \geq 0$, and

$$\text{Let } Ax + Iy = b \quad (5.4)$$

$$x_j = v_j^2 \quad j = 1, \dots, m \quad (5.5)$$

$$v_i = u_i^2 \quad i = 1, \dots, n \quad (5.6)$$

so that each x and each y has a positive value, since only real valued variables are considered. When the slack variables are included, their coefficients in the objective function are zero, so that the objective function is

$$w(x, y) = c'x + x'Rx + 0'y \quad (5.7)$$

where $0'$ is a row vector of zeroes for the slack variables.

To optimize (5.7) subject to (5.4) (5.6), $2n + m$ Lagrangian multipliers must be introduced. Let

$$\lambda' = [\lambda_1, \lambda_2, \dots, \lambda_n] \quad \phi' = [\phi_1, \phi_2, \dots, \phi_m] \quad \rho' = [\rho_1, \rho_2, \dots, \rho_m]$$

be Lagrangian multipliers so that the Lagrangian function for this problem can be formed. The Lagrangian function is

$$\begin{aligned}
 L &= c'x + x'Rx + 0'y + \lambda'(Ax + Iy - b) \\
 &\quad + \sum_{j=1}^m \rho_j(v_j - v_j^2) + \sum_{i=1}^n \phi_i(v_i - u_i^2)
 \end{aligned}$$

5

QUADRATIC PROGRAMMING

or

249

$$\begin{aligned}
 L &= \sum_{j=1}^m c_j x_j + \sum_{j=1}^m \sum_{k=1}^m r_{jk} x_j x_k \\
 &\quad + \sum_{i=1}^n \sum_{j=1}^m \lambda_i (a_{ij} x_j + y_i - b_i) + \sum_{j=1}^m \rho_j (x_j - v_j^2) \\
 &\quad + \sum_{i=1}^n \phi_i (y_i - u_i^2)
 \end{aligned}$$

The first-order conditions for maximizing or minimizing $w(x)$ in (5.1) are the same. They are determined by setting the $4n + 3m$ possible first derivatives equal to zero. Thus

$$\frac{\partial L}{\partial x_j} = c_j + 2 \sum_{k=1}^m r_{jk} x_k + \sum_{i=1}^n \lambda_i a_{ij} + \rho_j = 0 \quad j = 1, \dots, m \quad (5.8)$$

$$\frac{\partial L}{\partial y_i} = \lambda_i + \phi_i = 0 \quad i = 1, \dots, n \quad (5.9)$$

$$\frac{\partial L}{\partial v_j} = -2\rho_j v_j = 0 \quad \rho_j v_j^2 = 0 \quad j = 1, \dots, m \quad (5.10)$$

$$\frac{\partial L}{\partial u_i} = -2\phi_i u_i = 0 \quad \phi_i u_i^2 = 0 \quad i = 1, \dots, n \quad (5.11)$$

$$\frac{\partial L}{\partial \lambda_i} = \sum_{j=1}^m a_{ij} x_j + y_i - b_i = 0 \quad i = 1, \dots, n \quad (5.12)$$

$$\frac{\partial L}{\partial \rho_j} = x_j - v_j^2 = 0 \quad j = 1, \dots, m \quad (5.13)$$

$$\frac{\partial L}{\partial \phi_i} = y_i - u_i^2 = 0 \quad i = 1, \dots, n \quad (5.14)$$

Together, (5.10) and (5.13) imply

$$\rho_j v_j = 0 \quad j = 1, \dots, m \quad (5.15)$$

and (5.9), (5.11), and (5.14) provide

$$\lambda_i u_i = 0 \quad i = 1, \dots, n \quad (5.16)$$

The results of (5.8) and (5.12) are $n + m$ linear equations in $2n + 2m$ unknowns ($n\lambda$'s, $n y$'s, $m x$'s, and $m \rho$'s). However, $n + m$ of these variables have a zero value from (5.15) and (5.16). Thus (5.8), (5.12)

280

5

(5.15), and (5.16) provide 2^{n+m} sets of $n + m$ linear equations in $n + m$ variables. The results of (5.8), (5.12), (5.15), and (5.16) are the Kuhn-Tucker conditions for quadratic programming. These conditions can be summarized as

$$\begin{bmatrix} 2R & A^* & 1 & 0 \\ A & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ \lambda \\ \rho \\ y \end{bmatrix} = \begin{bmatrix} -c \\ -b \end{bmatrix} \quad (5.17)$$

where (5.15) and (5.16) are required as side conditions.

To determine the optimal solution, each alternative solution is substituted into the objective function. The one that optimizes $w(x)$ is then identified from among the feasible alternatives. For a specific case, it is often best to form a table, as has been done for the complete description method of linear programming.

The first-order conditions in (5.15), (5.17) are some of the necessary conditions to solve the problem delineated by (5.1)–(5.3). The first-order and second-order conditions are sufficient to solve the problem.

Properties of the Objective Function and Second-Order Conditions

The objective function $w(x) = c'x + x'Rx$ may either be convex or concave in x . Since a linear function is both convex and concave, the convexity or concavity of $w(x)$ depends solely on $x'Rx$; the results in Appendix B may be utilized to assess the convexity of this term. If $x'Rx$ is convex and $w(x)$ is to be minimized or if $x'Rx$ is concave and $w(x)$ is to be maximized, (5.1)–(5.3) constitute a convex programming problem; then the Kuhn-Tucker conditions are also sufficient for a global optimizing point x^* . Therefore, the second-order conditions for global optimality in the quadratic programming model are convexity of $x'Rx$ when minimizing $w(x)$ and concavity of $x'Rx$ when maximizing $w(x)$. These conditions are equivalent to the quadratic form $x'Rx$ being positive definite or positive semidefinite when $w(x)$ is to be minimized and its being negative definite or negative semidefinite when $w(x)$ is to be maximized (see Section 5.3). If these conditions are not satisfied, an optimizing point x^* satisfying (5.8), (5.12), (5.15), and (5.16) may not provide a global optimum.

Example 5.5

Consider the quadratic programming problem

$$\text{maximize } w = 20x_1 - 6x_1^2 + 40x_2 - 8x_2^2 + 30x_3 - 15x_3^2 \quad (5.18)$$

QUADRATIC PROGRAMMING

281

$$\begin{aligned} \text{subject to } & 3x_1 + 2x_2 + 6x_3 \leq 60 \\ & x_1, x_2, x_3 \geq 0 \end{aligned} \quad \begin{aligned} (5.19) \\ (5.20) \end{aligned}$$

In the matrix notation used for (5.1)–(5.3), the problem is to

$$\begin{aligned} \text{maximize } & w = c'x + x'Rx \\ \text{subject to } & Ax \leq b \\ & x \geq 0 \end{aligned}$$

where

$$c' = [20 \quad 40 \quad 30]$$

$$A = [3 \quad 2 \quad 6]$$

$$b = [60]$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad R = \begin{bmatrix} -6 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & -15 \end{bmatrix}$$

The Lagrangian function for this problem is

$$\begin{aligned} L = & 20x_1 - 6x_1^2 + 40x_2 - 8x_2^2 + 30x_3 - 15x_3^2 \\ & + \lambda(3x_1 + 2x_2 + 6x_3 + y - 60) + \sum_{j=1}^3 \phi_j(x_j - x_j^2) \\ & + \pi(y - y^2) \end{aligned}$$

This problem includes five Lagrangian multipliers λ , ϕ_1 , ϕ_2 , ϕ_3 , and π . The four variables that must be nonnegative are x_1 , x_2 , x_3 , and y ; $x_j = x_j^2$, $j = 1, 2, 3$, and $y = y^2$.

This problem has sixteen possible cases and solutions. These are determined by Kuhn-Tucker conditions, which follow for this problem.

$$\begin{bmatrix} -12 & 0 & 0 & 3 & 1 & 0 & 0 & 0 \\ 0 & -16 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & -30 & 6 & 0 & 0 & 1 & 0 \\ 3 & 2 & 6 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda \\ \phi_1 \\ \phi_2 \\ \phi_3 \\ y \end{bmatrix} = \begin{bmatrix} -20 \\ -40 \\ -30 \\ 60 \end{bmatrix}$$

252

5

The reader should compare these conditions with the general results provided by (5.15)–(5.17). The alternative solutions to this problem are given in Table 5.3. The optimal decision is that of Case 2.

Table 5.3. Alternative Solutions

Case	x_1	x_2	x_3	λ	ϕ_1	ϕ_2	ϕ_3	$\bar{z}$	w
1	5.667	5.0	5.0	20.0	0	0	0	0	-358.333
2	1.667	2.5	1.0	0	0	0	0	44.0	84.667
3	14.167	8.75	0	50.0	0	0	0	0	-1183.333
4	1.667	2.5	0	0	0	0	-330.0	0	66.667
5	7.667	0	6.026	25.128	0	0	-30.0	50.0	-583.974
6	1.667	0	1.0	0	0	-90.256	0	0	31.667
7	20.0	0	0	0	0	-40.0	0	49.0	-1999.927
8	1.667	0	0	73.333	0	-186.667	-469.999	0	16.667
9	0	6.724	7.754	33.793	-121.379	0	-30.0	55.0	-562.931
10	0	2.5	1.0	0	-20.0	0	0	0	65.0
11	0	30.0	0	220.0	-650.0	0	0	49.0	-6000.0
12	0	2.5	0	0	-20.0	0	-130.0	0	50.0
13	0	0	10.0	45.0	-155.0	-130.0	0	0	-11.999
14	0	0	1.0	0	-20.0	-40.0	0	54.0	15.0
15	0	0	0	60.0	-20.0	-40.0	-30.0	0	0
16	0	0	0	0	-20.0	-40.0	-30.0	60.0	0

This problem could have been solved more efficiently by implementing the Lagrangian algorithm delineated in Chapter 4. However, it is instructive to see the solutions for all 16 cases as illustrated in Table 5.3. Notice that only a small portion of these cases would be solved by the Lagrangian algorithm before locating the global solution (Case 2).

For linear models, the Lagrangian multipliers have been shown to be the dual variables. Together, the optimal allocations for the primal and dual linear programming problems identify a saddle point. In quadratic programming this same result is true; however, this feature does not generate further considerations such as a simple game. The interpretation of the Lagrangian multipliers for the quadratic problem is very similar to that for the linear case. Considering (5.9), (5.15), and (5.16), the key Lagrangian multipliers are the elements of λ . Either λ_i or y_i is zero. Since λ_i is the premultiplier of the linear constraint having the slack variable y_i , the interpretation for the linear programming model can be extended to the primal quadratic problem. The dual models are described in Section 5.6.

QUADRATIC PROGRAMMING

253

5.5 WOLFE'S SIMPLEX METHOD FOR QUADRATIC PROGRAMMING

A vast number of algorithms have been presented in the relevant literature of quadratic programming. Many have been developed to deal with problems having special features. The short form of the Wolfe algorithm [1959] may be applied to any quadratic programming problem in which the necessary second-order conditions are satisfied.

The Kuhn-Tucker conditions for quadratic programming are given by (5.15)–(5.17) or

$$\begin{aligned} 2R\mathbf{x} + \mathbf{A}'\lambda + \mathbf{I}\rho &= -\mathbf{c} \\ \mathbf{A}\mathbf{x} + \mathbf{I}\mathbf{y} &= \mathbf{b} \end{aligned} \quad (5.21)$$

For any constraint among

$$\sum_{j=1}^n a_{ij}x_j + y_i = b_i \quad i = 1, \dots, n$$

where b_i is negative, the constraint is to be multiplied by minus one. The form $\mathbf{A}\mathbf{x} + \mathbf{I}\mathbf{y} = \mathbf{b}$ will be employed throughout this section, assuming that each b_i is positive but that the coefficient of y_i could be negative.

The Framework of the Algorithm

The short form of the Wolfe algorithm is a two-phase linear programming process. An artificial variable with a coefficient of plus one is added to each constraint of $\mathbf{A}\mathbf{x} + \mathbf{I}\mathbf{y} = \mathbf{b}$. Let $\hat{\mathbf{x}}$ denote an $n \times 1$ vector of artificial variables so that $\mathbf{I}\hat{\mathbf{x}}$ is a diagonal matrix. Then each constraint of the form $\mathbf{A}\mathbf{x} + \mathbf{I}\mathbf{y} + \mathbf{I}\hat{\mathbf{x}} = \mathbf{b}$ has a slack variable y_i and an artificial variable, $\hat{x}_i$. At least one of y_i or $\hat{x}_i$ has a coefficient of plus one.

If the i th linear constraint is

$$\sum_{j=1}^n a_{ij}x_j \geq b_i \quad b_i > 0$$

the artificial variable will be needed for the initial basic feasible solution. Inequalities of this sort are the usual occurrence when a quadratic minimization problem is considered.

Two artificial variables π_i^* and π_j^* will be included in each condition among those in $2R\mathbf{x} + \mathbf{A}'\lambda + \mathbf{I}\rho = -\mathbf{c}$. Let π^* and π^* be two-